PROJECT BRIEF / 01

Payment Status & Reconciliation Analysis

Initial product requirements document | Independent portfolio project

Prepared for	Project owner	Version / date
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Current stage: initiation and requirements drafting

This document is an initial planning work sample prepared with AI assistance. It records a proposed scope, business rules, and validation approach. Dataset creation, SQL implementation, dashboard development, and testing have not yet been completed. It is not evidence of a deployed product or achieved business results.

1. Problem and proposed solution

In a simulated small-store setting, an order, its payment attempt, a refund, and a bank settlement represent different records. Without a consistent view, an operator may confuse a pending payment with a failure, overlook a refund, or miss a settlement discrepancy.

Build a small analytical prototype that links these records, reports payment outcomes, and creates an exception queue for investigation. The prototype will use synthetic INR transactions only.

2. Intended users and value

Store operator: identify orders whose payment needs investigation.
Finance / operations analyst: compare expected and recorded settlement amounts.
Product analyst: understand outcome patterns and recommend workflow improvements.

3. Scope boundaries

Included: payment success, failure and pending status; payment attempts; full and partial refunds; settlement matching; SQL analysis; a basic dashboard; requirements and UAT scenarios.
Excluded: real money movement, live gateways, production integrations, customer personal data, automated refund execution, chargebacks, and regulatory certification.

Workflow & requirements

4. Simplified workflow and assumptions

Order created → payment attempt recorded → status updated → optional refund recorded → settlement compared → exceptions reviewed.

For this prototype, "success" means a captured payment. Each retry has a new attempt ID. Payment outcome, refund status, and settlement status are stored separately. Actual provider definitions and settlement arrangements may differ. A failed attempt may be followed by a successful retry; a later update may resolve a pending attempt.

ID	Requirement / user story	Acceptance criteria
R1	As an operator, I can inspect each payment attempt and its latest status.	Store unique attempt IDs, order links, event timestamps and as-of time. Replayed event IDs do not create duplicate records.
R2	As an analyst, I can identify pending payments requiring review.	Flag pending attempts older than a configurable 24-hour demo threshold. Do not automatically mark them failed.
R3	As an operator, I can distinguish successful, pending and failed refunds.	Track refunds independently. Only successful refunds reduce the refundable balance; total successful refunds cannot exceed captured value.
R4	As a finance analyst, I can identify settlement exceptions.	Match by settlement reference and currency; compare expected net and actual settled values. Missing, partial and amount-mismatch cases remain visible.
R5	As a product analyst, I can filter and compare payment outcomes.	Filter by attempt date, method and status. Display counts, INR amounts, metric definitions and snapshot time.

5. Proposed deliverables

A synthetic dataset and data dictionary; PostgreSQL schema and SQL queries; an Excel dashboard; a brief findings report; and a UAT checklist. These are planned deliverables, not completed outputs.

Data & analytical approach

6. Proposed data model

Table / grain	Core fields
orders / one order	order_id, created_at, order_amount_paise, currency
payment_attempts / one attempt	attempt_id, order_id, method, amount_paise, status, created_at, status_updated_at
payment_events / one event	event_id, attempt_id, status, occurred_at, received_at
refunds / one refund	refund_id, attempt_id, amount_paise, status, processed_at, settlement_batch_id
settlement_items / one batch line	line_id, batch_id, attempt_id or refund_id, line_type, signed_amount_paise
settlement_batches / one settlement	batch_id, due_at, settled_at, actual_net_paise, currency

Use integer paise for amounts and consistent timestamps. Each successful payment belongs to one expected settlement batch in this simplified model. Refunds may be assigned to a later batch. Settlement lines distinguish captures, refund deductions, fees and fee taxes. Missing actual values are not treated as zero.

7. Planned metrics and reconciliation rules

Attempt success rate: successful attempts / all initiated attempts in the selected creation-date cohort, measured as of the displayed snapshot. Pending attempts stay in the denominator.
Order paid rate: distinct orders with at least one successful attempt / distinct orders with an attempt. Report separately so retries do not inflate paid orders.
Refund value: sum of successful refund amounts, grouped by processing date; pending refunds shown separately.

Expected net by settlement batch: captures assigned to that batch minus successful refund deductions assigned to it, fees and fee taxes.
Variance: actual settled amount minus expected net. Exact paise matching is the demo rule. A batch is overdue only after its configured due time has passed and the actual settlement is missing.

8. SQL questions to answer

Which payment methods have the most failed attempts? Which pending attempts exceed the review threshold? Which orders have multiple successful attempts? Which refunds violate amount rules? Which due settlement batches are missing or have a non-zero variance?

Validation & next steps

9. Planned UAT scenarios

Scenario	Expected outcome
Failed attempt followed by success	Two attempts; one paid order. Failure remains in attempt-level reporting.
Pending attempt, then late success	Flag before resolution; remove pending flag after valid later event. Retain event history.
Same event received twice	One event retained; counts and amounts do not increase.
Two successful attempts for one order	Order counted once as paid; duplicate-payment exception raised for review.
Partial refunds of INR 300 and INR 200 on INR 1,000 capture	Successful refunded value is INR 500. Pending or failed refunds do not reduce the remaining balance.
INR 1,000 capture less INR 100 refund, INR 20 fees and INR 3.60 fee tax	Expected net INR 876.40. If actual is INR 870, flag variance of minus INR 6.40. These are invented test values.
Settlement due tomorrow, not yet received	Awaiting settlement; not overdue. Missing settlement after due time becomes an overdue exception.

10. Proposed milestones

Next: review this draft, confirm assumptions and create the data dictionary.
Then: generate approximately 1,000 synthetic attempts with deliberately seeded exceptions and implement SQL.
Then: build the dashboard, execute UAT and write findings.
Before publishing results: retain reproducible queries, document limitations and distinguish simulated outcomes from real business impact.

11. Definition of done

All listed scenarios pass; duplicate and missing records are handled explicitly; dashboard totals reconcile to SQL; seeded exceptions are detected; metric denominators and dates are documented; and a reviewer can reproduce the analysis from the synthetic source files.

Portfolio statement

Independent payments-analysis project initiated by Harsh Sagar. Current evidence: an initial PRD and proposed validation plan. Implementation and analytical results are pending. No client delivery, live integration or revenue improvement is claimed.